

AmazonHelp AI Support Agent

Hiver SDE Intern Take-Home Assignment

1. Problem

Build an AI support agent that classifies incoming customer messages, retrieves historical AmazonHelp resolutions, drafts grounded replies, and decides whether to auto-answer or escalate.

The system prioritizes safe evidence use and evaluation over unnecessary model complexity.

2. Dataset and evaluation

AmazonHelp was selected from the Twitter Customer Support corpus because its support history contains diverse customer issues and multi-turn interactions.

The final primary human benchmark contains 152 genuinely human-labeled examples.

Retrieval evaluation uses a temporal split and excludes Golden conversation roots from the retrieval corpus.

3. Intent taxonomy

The final taxonomy contains 11 support intents:

delivery_status, delivery_problem, order_issue, preorder_issue, returns_replacements, refund_issue, payment_billing, account_security, prime_membership, digital_product_support, general_support.

The final classifier combines high-precision boundary rules with a MiniLM semantic fallback.

4. Intent evaluation

The human Gold set is treated as a fixed benchmark rather than repeatedly tuning individual examples.

5. Retrieval

BM25, dense MiniLM retrieval, and hybrid retrieval were evaluated. Dense retrieval was selected because it performed better on stricter usefulness and actionability measures than the broader hybrid result.

Similarity scores are treated as ranking signals, not calibrated probabilities.

6. Grounding and response policy

Finding a related historical conversation is not by itself sufficient for automatic answering.

The final policy requires strong evidence and high actionability before an answer is considered safe to automate.

The response generator is grounded in historical AmazonHelp responses and is instructed not to invent unsupported policies, refunds, eligibility conditions, or commitments.

7. End-to-end results

The system auto-answered 6 cases and escalated 146 cases.

8. What is misleading about my headline number?

The most misleading headline number is the 98.0% grounded rate.

That number does not mean the agent can safely answer that percentage of requests.

Only 36.8% of cases had strong evidence and 42.1% had high actionability. The resulting automatic-answer rate was only 3.9%.

Therefore:

related retrieval != strong evidence != actionable evidence != safe auto-answer

This distinction is intentional and forms the main safety property of the system.

9. Human evaluation and limitations

Intent agreement between the system and the human Gold labels was 56.6%.

Escalation labels are highly imbalanced in the human benchmark, so escalation agreement is treated as diagnostic rather than a definitive production safety estimate.

The LLM judge is treated as an evaluation instrument rather than ground truth. A larger independent human review set would be appropriate for production validation.

10. Engineering decisions

The final system prioritizes:

- 1. leakage-safe temporal evaluation
- 2. Golden-root exclusion from retrieval
- 3. dense retrieval selected using stricter quality criteria
- 4. explicit intent boundaries
- 5. conservative evidence and actionability gating
- 6. escalation instead of unsupported generated claims

Repository

<https://github.com/gotnochill815-web/hiver-amazonhelp-agent>

Final metric snapshot

Metric	Value
Intent accuracy	56.6%
Intent macro F1	58.3%
Intent weighted F1	56.3%
Grounded rate	98.0%
Strong evidence	36.8%
High actionability	42.1%
Auto-answer rate	3.9%